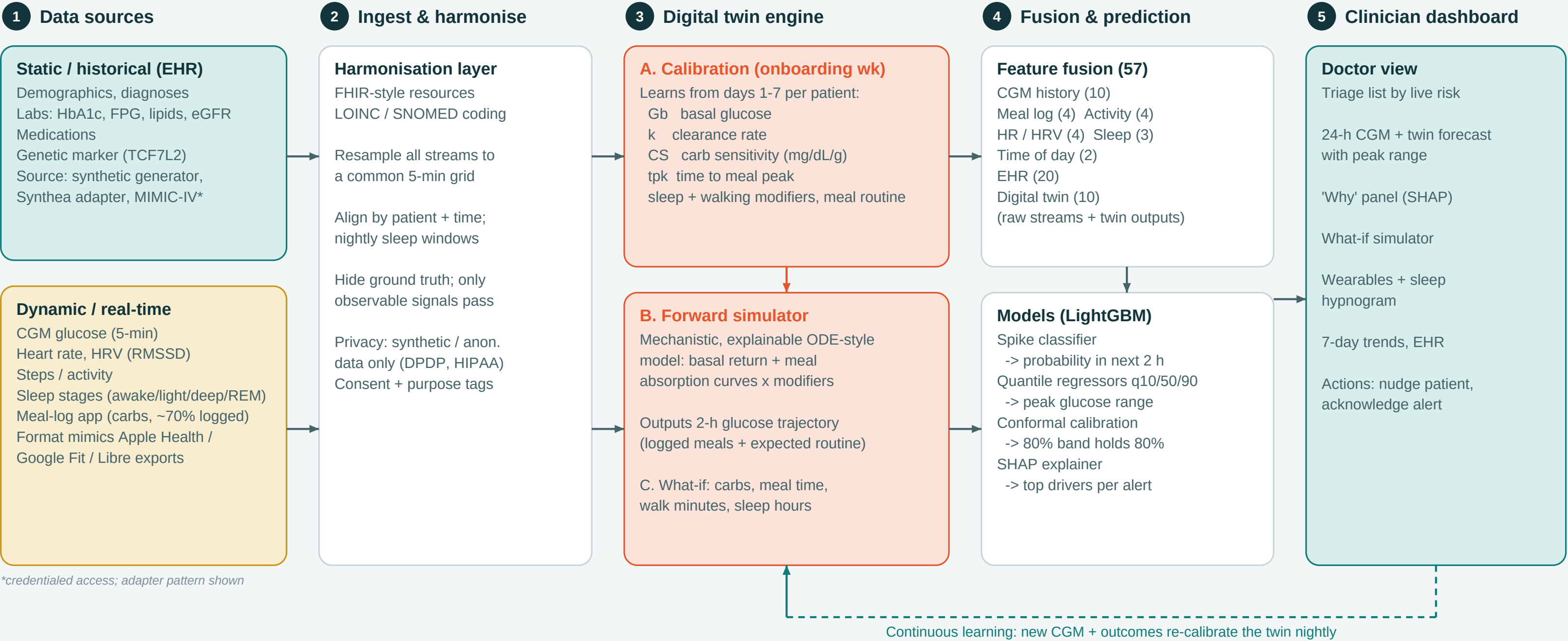


GlucoTwin: system architecture

Type 2 diabetes digital twin that fuses EHR + wearable streams to predict a glucose spike (>180 mg/dL) up to 2 hours ahead



Evaluation protocol

Patient-level split: test patients never seen in training (no leakage)
5-fold GroupKFold CV, ablation per stream, trend-arrow clinical baseline
Clinical metrics: detection rate, lead time, alert episodes/day, calibration

Production path (roadmap)

REST API (FastAPI) + message queue for device streams
ABDM / FHIR R4 integration for hospital EHRs; on-device CGM ingestion
Prospective validation with a clinical partner before any patient use

How one prediction is made (every 15 minutes)

Hybrid design: a physiology twin supplies interpretable structure; ML corrects what the equations miss

1 Read the last 2 hours

CGM level, slopes (15/30/60 min),
acceleration, variability
Steps, HR, HRV
Meals logged in last 4 h

2 Add context

Last night: sleep hours, deep %,
REM %, overnight HRV vs baseline
Time of day
EHR: HbA1c, BMI, meds, TCF7L2...

3 Run the twin

Simulate 2 h ahead with this
patient's calibrated parameters
Add expected meal from their
personal routine (probability-weighted)

4 Predict + explain

P(spike > 180 in 2 h)
Peak range q10-q90 (+ conformal)
SHAP top drivers
Alert if P >= threshold

Twin forward model

$$G(t) = G_b + R_0 * \exp(-k t) \\ + \text{SUM_meals} \quad CS * \text{carbs} * S * W * f(t + \text{ago})$$

$$f(\tau) = (\tau / \text{tpk}) * \exp(1 - \tau / \text{tpk})$$

$$S = 1 + \text{beta_sleep} * \max(0, 7 - \text{sleep_hours})$$

$$W = 1 - \text{walk_effect} * \min(\text{walk_min}, 30) / 30$$

$$R_0 = \text{unexplained current excess (decays)}$$

Parameters are fitted per patient from the onboarding week and shrunk
toward population priors when data are sparse (empirical-Bayes style).

Validation: twin-estimated CS vs hidden true insulin sensitivity, $r = -0.87$

Why hybrid, not pure deep learning?

- Works from 1 week of data per patient (small-data regime)
- Every parameter has clinical meaning a doctor can check
- Enables what-if simulation via a mechanistic core
- Adding the twin lifts CV AUROC 0.928 -> 0.934 over plain fusion
- Full model halves 2-h peak error vs persistence (24 -> 12 mg/dL)
- Gradient boosting + SHAP: fast on CPU, auditable, easy to deploy
- Deep sequence models (e.g. transformers) are a roadmap item
once real multi-month CGM data is available